# Innovation in Hypotheses

## Hypothesis 1 – Global Pollution Fingerprints of Cities

**Statement:**
Each city can be represented by a unique “pollution fingerprint” — a distinctive combination of pollutants that captures its dominant pollution sources (e.g., traffic, industrial, or photochemical smog). These fingerprints may form natural clusters that correspond more closely to geography and local climate than to national borders.

**Rationale:**
Even though cities are grouped by country, pollution patterns depend more on physical and socioeconomic factors. For example, cities at similar latitudes may have comparable heating and transport emissions. Since the dataset covers thousands of cities and 13 pollutants, it enables us to identify these latent patterns and see whether pollution types are driven more by geography than by political boundaries.

**Testing Approach:**

1. Aggregate pollutant data by *City* and *Pollutant* to compute average values.
2. Normalize pollutant levels (e.g., z-score per pollutant).
3. Apply clustering algorithms such as **K-Means** or **DBSCAN** to identify groups of cities with similar pollution profiles.
4. Evaluate cluster quality with silhouette scores.
5. Map the resulting clusters using latitude and longitude to visually analyze geographical relationships.

## Hypothesis 2 – Coastal Influence on Particulate Correlation

**Statement:**
The correlation between PM2.5 and PM10 concentrations is weaker in coastal cities compared to inland ones, due to sea salt particles that increase PM10 but not PM2.5.

**Rationale:**
Inland particulate matter mainly originates from combustion, which causes PM2.5 and PM10 to move together strongly. Coastal areas, however, have additional natural PM10 from sea spray, which decouples the relationship between fine and coarse particles. By using coordinates, we can differentiate coastal from inland cities and assess this environmental influence globally.

**Testing Approach:**

1. Derive a new binary variable "Coastal" (1 if within ~50 km of a coastline).
2. For each group (Coastal vs. Inland), calculate the **Pearson correlation coefficient** between PM2.5 and PM10.
3. Use a **Fisher r-to-z test** to check if the correlation difference is statistically significant.
4. Visualize with scatter plots for each group and a bar chart comparing correlation strengths.

## Hypothesis 3 – Diurnal Pollution Patterns and City Behavior

**Statement:**

Pollution levels follow daily cycles reflecting human activity — with $NO_2$ and CO peaking during morning and evening rush hours, and $O_3$ peaking in the afternoon due to sunlight-driven reactions. The strength and timing of these cycles vary with city size and industrial activity.

**Rationale:**

Hourly readings in the dataset allow us to uncover how pollution mirrors daily human behavior. Larger or more industrial cities might show sharper $NO_2$ peaks, while smaller or residential areas display flatter cycles. Identifying these "pollution rhythms" can reveal the human and industrial dynamics that shape air quality.

**Testing Approach:**

1. Extract the **hour** from the *Last Updated* field.
2. Group data by *City*, *Pollutant*, and *Hour*, then calculate mean hourly pollutant levels.
3. Plot 24-hour cycles for $NO_2$, CO, and $O_3$.
4. Measure amplitude (max–min) of pollutant variation per city.
5. Regress amplitude values against proxies of city size (number of monitoring stations, etc.) to test how diurnal variation scales with urban intensity.

## Hypothesis 4 – Trans-Boundary Pollution Connectivity

**Statement:**

Cities located near international borders have more similar pollution levels to their cross-border neighbors than to cities within their own country, showing the transnational spread of air pollution.

**Rationale:**
Air pollutants travel with wind and weather systems, ignoring political boundaries. For instance, pollution between border cities such as Delhi–Lahore or Vienna–Bratislava may be more similar than between distant cities within the same nation. The dataset's geographical coordinates make it possible to test this empirically.

**Testing Approach:**

1. Compute average pollutant values per *City*.
2. Calculate geographic distances between all city pairs using the **Haversine formula**.
3. Identify each city's 10 nearest neighbors.
4. Label neighbor pairs as "Cross-border" or "Same-country."
5. Compute Pearson correlation of pollutant averages between each city pair.
6. Compare correlation distributions using a **Mann–Whitney U test** to see if cross-border pairs are significantly more similar.

## Hypothesis 5 – Network-Level Sensor Bias and Volatility

**Statement:**
Different monitoring networks (Source Names) display systematic differences in average readings and variability, even when measuring the same pollutant in the same city — suggesting calibration or sensor-type biases.

**Rationale:**
Your dataset includes over 100 sources — from government stations to community sensors — that may not share calibration standards. Low-cost sensors tend to have higher noise or overestimate pollutants compared to reference monitors. Quantifying this inconsistency is key for improving data reliability and comparability.

**Testing Approach:**

1. Within each city and pollutant, group data by *Source Name*.
2. Compute mean and variance of pollutant *Value* for each source.
3. Use **ANOVA** to test mean differences and **Levene's test** for equality of variances.
4. Plot boxplots or violin plots showing distributions by source.
5. Identify and discuss networks that systematically deviate from others.